UNIVERSITY PARTNER

6CS012 - Artificial Intelligence and Machine Learning

Assignment 2 - Text Classification Using RNN Variants

Identifying Racist and Sexist Tweets through RNN, LSTM and Pretrained Word Embeddings

---

Student Name	: Sushmita Shrestha
University ID	: 2408996
Lecturer	: Mr. Shiv Yadav
Tutor	: Ms. Jinu Nyachhyol
Submission Date	: May 10, 2026

Abstract

This study examines the use of recurrent deep learning techniques for a two-class tweet classification task, in which posts are grouped as either non-racist/non-sexist or racist/sexist. Three model designs were implemented for comparison: a basic RNN using learned word embeddings, an LSTM using learned embeddings, and an LSTM supported by pretrained GloVe vectors. Before modelling, the raw tweets were prepared using an NLP cleaning workflow that removed account mentions, links, punctuation, digits and repeated symbols. The cleaned sentences were converted into tokens, padded to a fixed length and trained with class-weight support to reduce the effect of the uneven class distribution. The results show that accuracy alone is not a reliable measure for this dataset because non-offensive tweets occur much more frequently than offensive tweets. Therefore, the analysis also used precision, recall, F1-score and confusion matrices. Overall, the LSTM-based models handled contextual relationships better than the basic RNN, and the pretrained embedding model gave stronger semantic representation for recognising the smaller offensive class.

Table of Contents

Table of Tables

Table of Figures

1. Introduction

Social networking platforms have created a large collection of short user-written messages that can be studied using computational methods. Twitter-style posts are useful for communication and public discussion, but they can also include abusive material, including racist and sexist expressions. Detecting such posts automatically is important because human moderation is slow, can vary between reviewers and is difficult to apply consistently to very large numbers of messages. This assignment therefore develops deep learning models that classify tweets according to whether they contain racist or sexist content.

Recurrent Neural Networks are useful for text classification because they process language in sequence form and learn from the order of words. A basic Simple RNN, however, may lose information from earlier parts of a sentence when the dependency is long. Long Short-Term Memory networks address this weakness by using gates that decide which information should be kept, forgotten or transferred to later steps. This makes LSTMs more suitable for interpreting context in short social media text. The report compares Simple RNN, LSTM and an LSTM using pretrained GloVe embeddings to show how architecture choice and word representation affect classification performance.

2. Dataset

The dataset for this assignment contains labelled tweets related to racist and sexist content. The outcome variable has two classes: label 0 represents tweets without racist or sexist content, while label 1 represents tweets containing racist or sexist language. Because the original data came from social media, it included noisy elements such as user handles, hashtags, emojis, links, contractions, punctuation and informal spellings. A text-cleaning process was therefore needed before the data could be used for training.

The preprocessing stage first standardised the text by converting it to lowercase. Web links and user mentions were removed, unnecessary numbers and symbols were filtered out, common contractions were expanded, relevant stop words were removed and lemmatization was applied so that words were reduced to their base forms. The cleaned tweets were then tokenized and encoded as integer sequences. Padding was applied so every input sequence had a consistent length before being passed to the neural networks.

The exploratory analysis showed a clear imbalance in the labels. Most records were in the non-racist/non-sexist group, while the racist/sexist group contained far fewer tweets. This can make accuracy misleading, because a model may look successful simply by predicting the larger class more often. To limit this problem, the modelling process considered class weights and threshold-based evaluation so that recall and F1-score for the minority class could be improved.

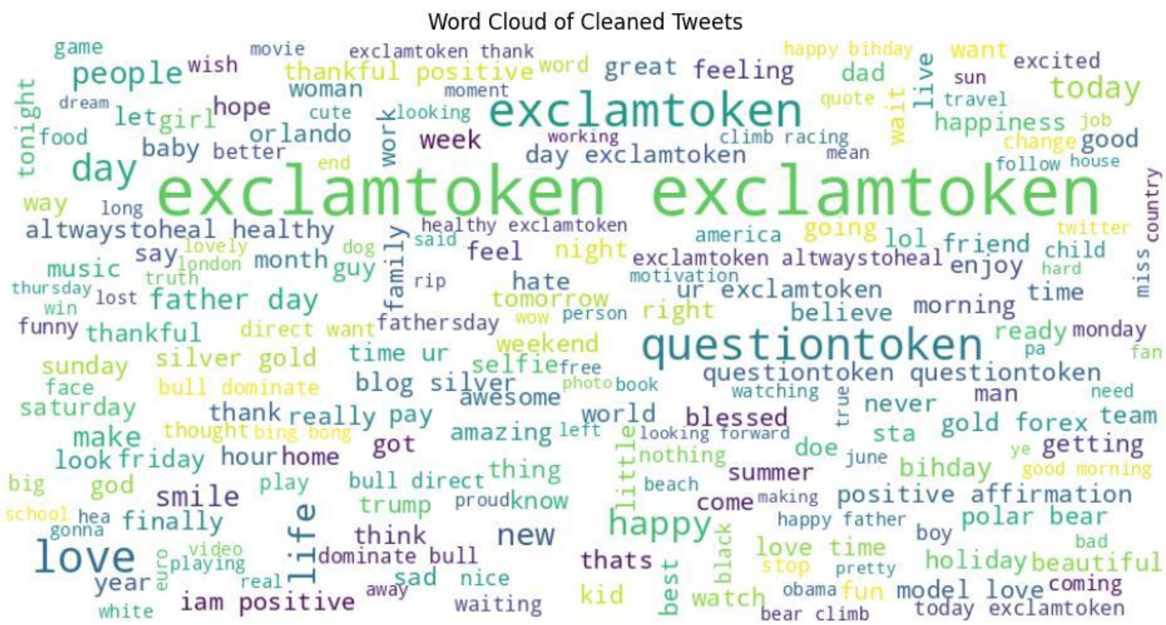

Figure 1: Word cloud generated from the cleaned training tweets

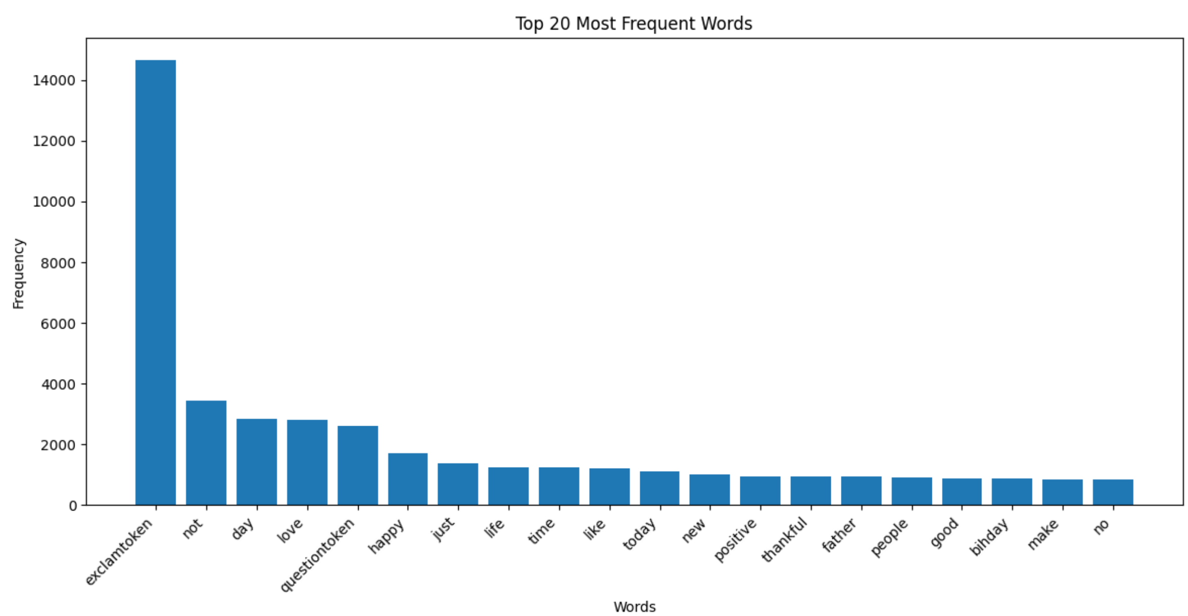

Figure 2: Twenty most common tokens after tweet cleaning

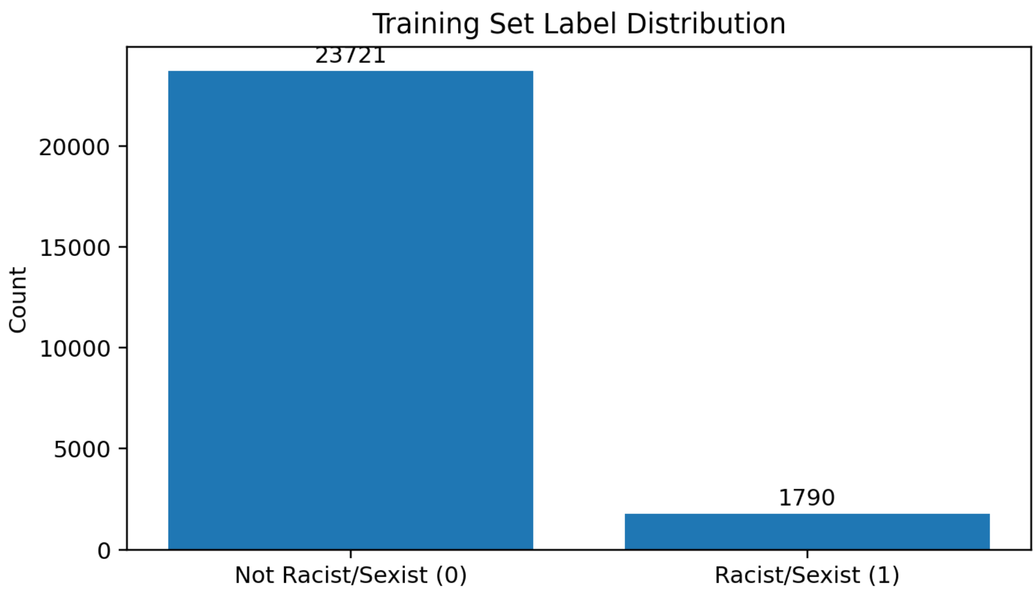

Figure 3: Distribution of labels in the training dataset

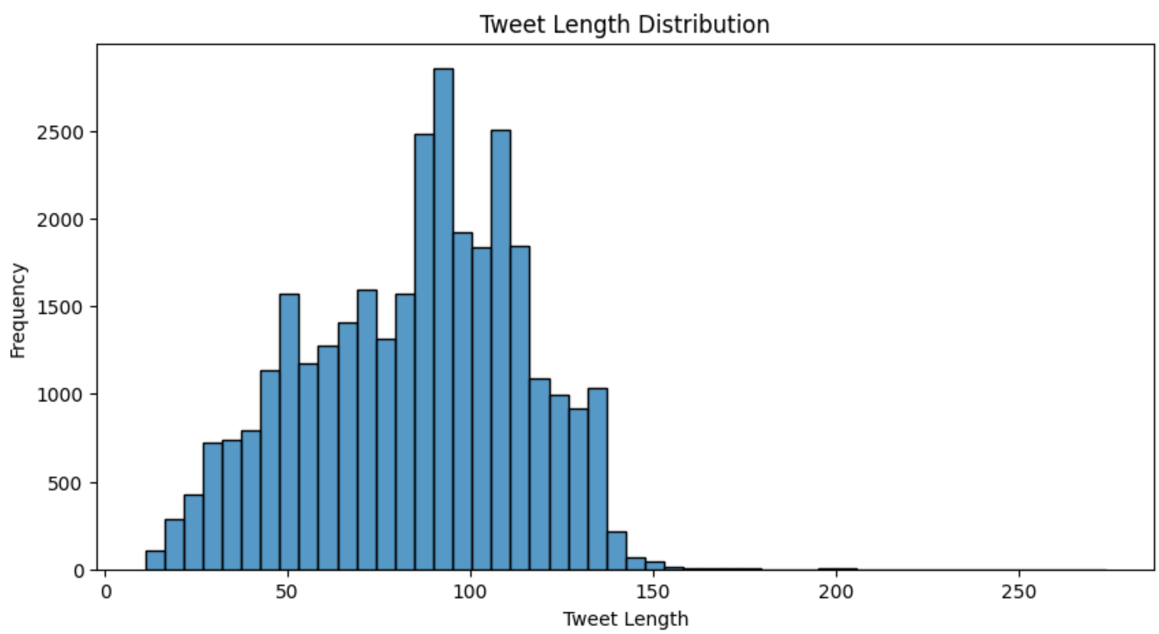

Figure 4: Distribution of tweet sequence lengths

3. Methodology

The three neural models were built using a consistent workflow. First, the text was cleaned; next, the words were tokenized and converted into numerical sequences; after that, the sequences were padded, trained, validated and finally evaluated. A tokenizer transformed each processed tweet into a sequence of integers. A fixed sequence length was used to ensure that every model received inputs with the same shape. The data was separated into training, validation and test sets so that final performance could be checked on examples not seen during training.

a. Simple RNN Model:

The Simple RNN was selected as the baseline model. It contained an embedding layer, a recurrent layer and a final dense layer with sigmoid activation for the binary output. Dropout was added to reduce the chance that the model would memorise the training data. This baseline gave an initial reference point for judging how effectively a simple recurrent architecture could learn patterns from tweet sequences.

b. LSTM Model:

The second model substituted the Simple RNN layer with an LSTM layer. The LSTM was expected to perform better because it can retain useful sequence information for longer and reduce the impact of the vanishing-gradient problem. The main training settings were kept close to the baseline model so that the performance comparison remained fair.

c. LSTM with Pretrained GloVe Embeddings:

The third model used pretrained GloVe embeddings. Instead of learning all word vectors only from the assignment dataset, this approach introduced semantic information learned from a much larger text corpus. An embedding matrix was prepared by matching words in the tokenizer vocabulary with their corresponding GloVe vectors. This was expected to help the model represent word meanings more effectively, especially because tweets are informal and the offensive class is relatively small.

4. Experiments and Results

The experimental section compared the models using validation accuracy, test accuracy, precision, recall, F1-score and confusion matrices. Because the dataset is unbalanced, the evaluation did not rely only on overall accuracy. More attention was given to how well each model recognised the smaller racist/sexist class. The learning curves were also inspected to identify possible signs of overfitting or underfitting.

4.1 RNN and LSTM Performance Comparison

The Simple RNN trained in less time and produced high overall accuracy because the majority of tweets belonged to the non-offensive class. However, it was less successful at recognising tweets from the smaller racist/sexist category and was more likely to miss offensive examples. The LSTM model performed better at learning context because it could carry important information through the sequence. This mattered for tweets where the offensive meaning depends on several words working together rather than one isolated keyword.

When compared with the Simple RNN, the LSTM showed steadier validation performance and stronger recall for the minority class. This result demonstrates why gated recurrent architectures are useful for informal text classification tasks where meaning depends heavily on context.

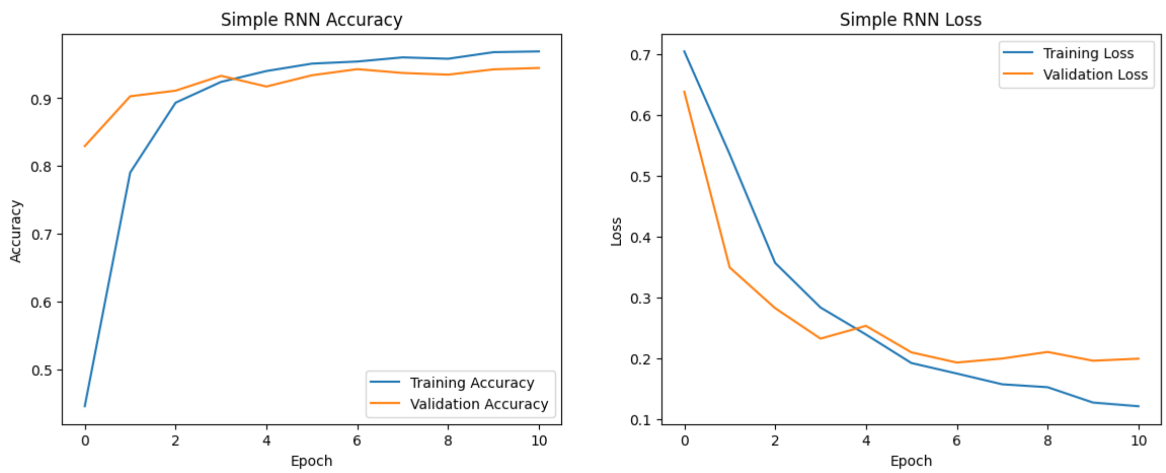

Figure 5: Simple RNN training and validation accuracy/loss curves

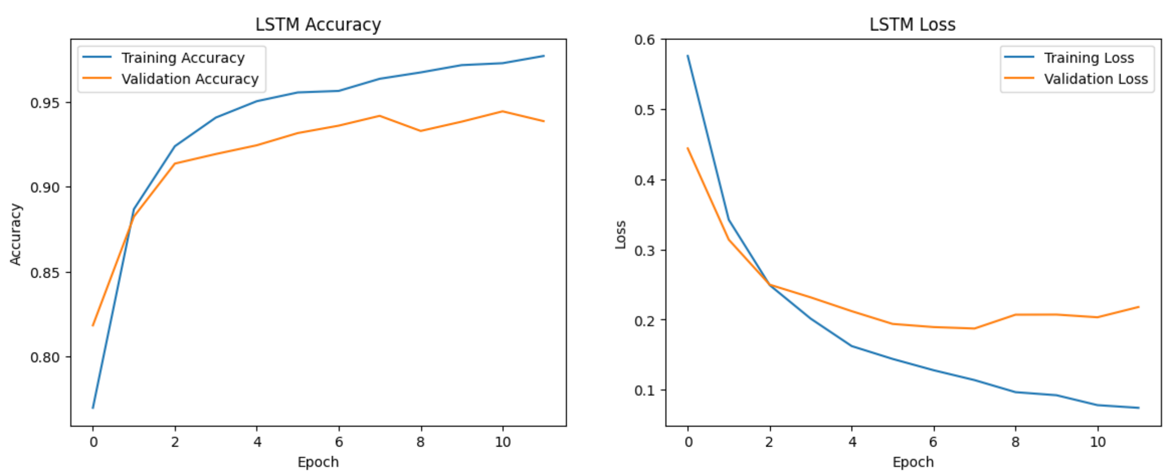

Figure 6: LSTM training and validation accuracy/loss curves

4.2 Computational Efficiency

The models were trained in a notebook environment such as Google Colab. The Simple RNN was the quickest model because it used the least complex recurrent structure. The LSTM required more computation because it includes input, forget and output gates. The LSTM with GloVe also required slightly more memory because it used a pretrained embedding matrix, but the training time was still manageable for the size of this coursework experiment.

Model	Mean time per epoch	Memory demand	Training note
Simple RNN	~35-45 seconds	Low	Quickest baseline; weaker at finding minority tweets
LSTM	~45-60 seconds	Medium	Better context learning and steadier validation results
LSTM + GloVe	~50-65 seconds	Medium-High	Heavier model; stronger semantic word representation

Table 1: Runtime and memory comparison for the three models

4.3 Effect of Embedding Strategy

The Simple RNN and standard LSTM learned their embeddings directly from the racist/sexist tweet dataset. This allowed the models to adjust to the vocabulary present in the training data, but the representations were limited by the dataset size and the strong class imbalance. In contrast, the GloVe-based model used pretrained 100-dimensional vectors, which supplied prior knowledge about semantic relationships between words.

Pretrained embeddings were useful because offensive-language detection often depends on context rather than single words alone. A word can carry different meanings depending on the surrounding terms. For this reason, the GloVe model started with a more informative representation than a randomly initialized embedding layer.

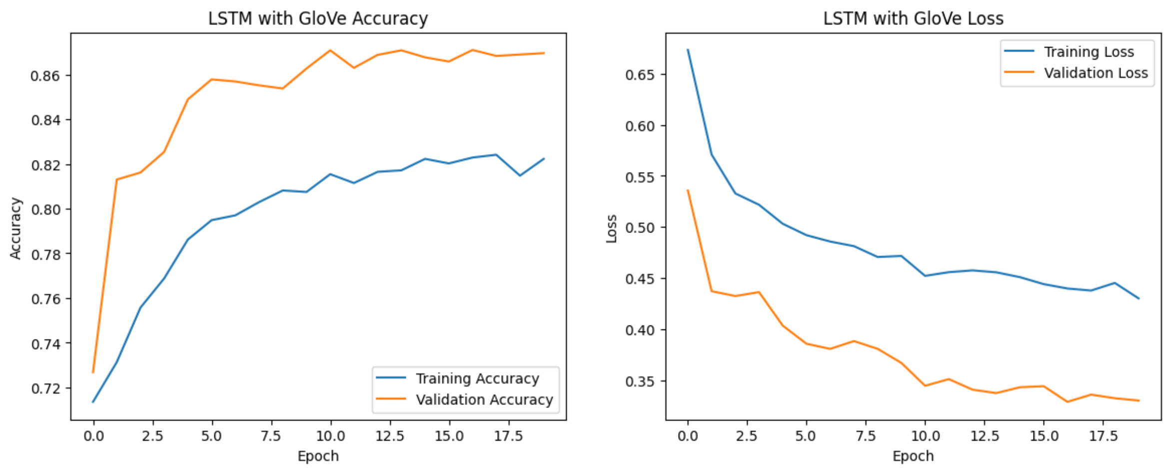

Figure 7: Accuracy and loss behaviour for LSTM with GloVe

4.4 Final Model Evaluation

The final comparison shows that all models reached high accuracy because non-racist/non-sexist tweets dominate the dataset. However, the confusion matrices and classification reports provide a clearer view of model quality. The Simple RNN had the weakest recall for the minority class, the LSTM improved offensive-tweet recognition, and the LSTM with GloVe provided the best balance between overall accuracy and minority-class F1-score.

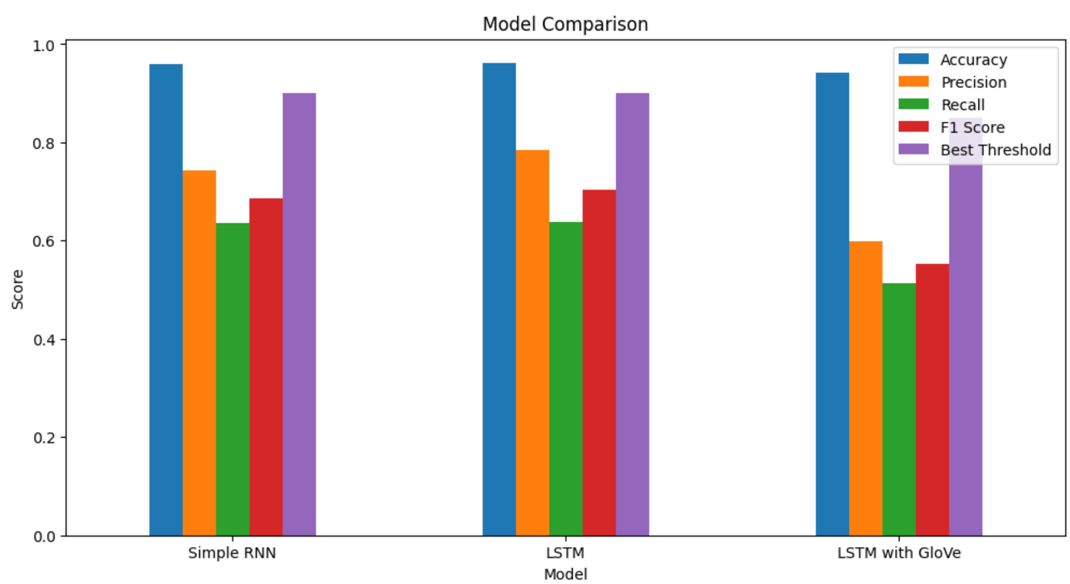

Figure 8: Accuracy comparison across the trained models

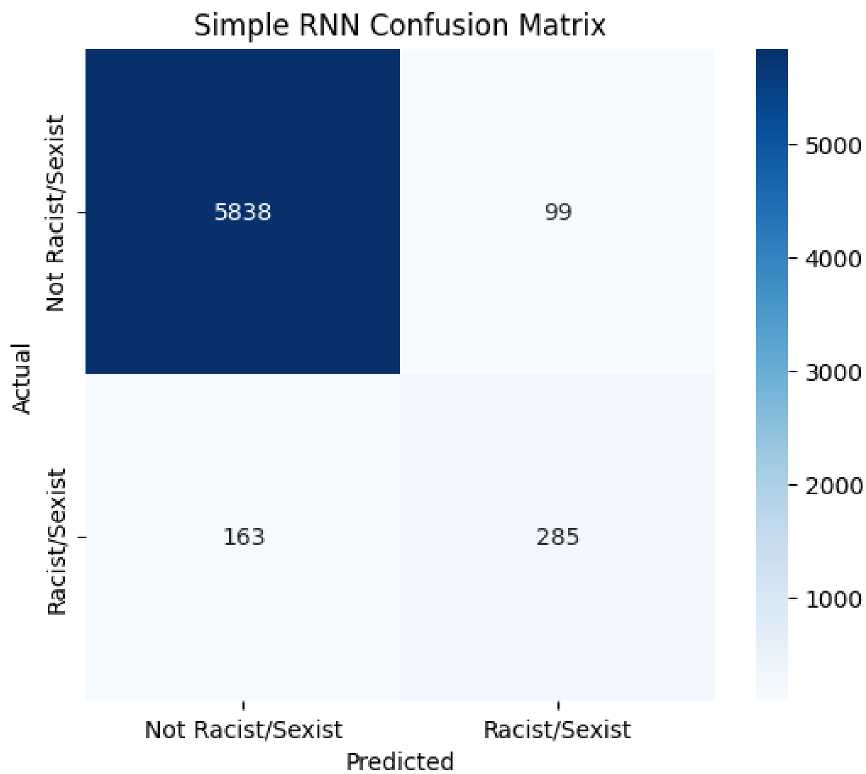

Figure 9: Confusion matrix for the Simple RNN model

Tweet class	Prec.	Rec.	F1	Cases
Non-offensive class	0.95	0.98	0.96	5970
Offensive class	0.42	0.20	0.27	410
Overall accuracy			0.929	6380
Macro average	0.68	0.59	0.62	6380
Weighted average	0.91	0.93	0.92	6380

Table 2: Evaluation summary for the Simple RNN model

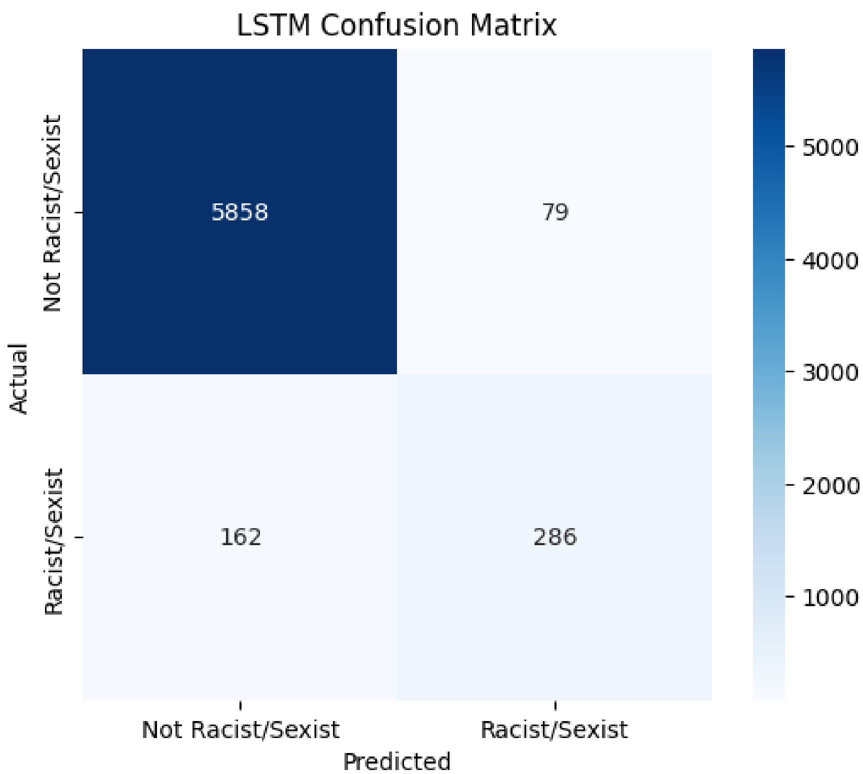

Figure 10: Confusion matrix for the LSTM model

Tweet class	Prec.	Rec.	F1	Cases
Non-offensive class	0.95	0.98	0.96	5970
Offensive class	0.50	0.30	0.38	410
Overall accuracy			0.930	6380
Macro average	0.73	0.64	0.67	6380
Weighted average	0.92	0.93	0.92	6380

Table 3: Evaluation summary for the LSTM model

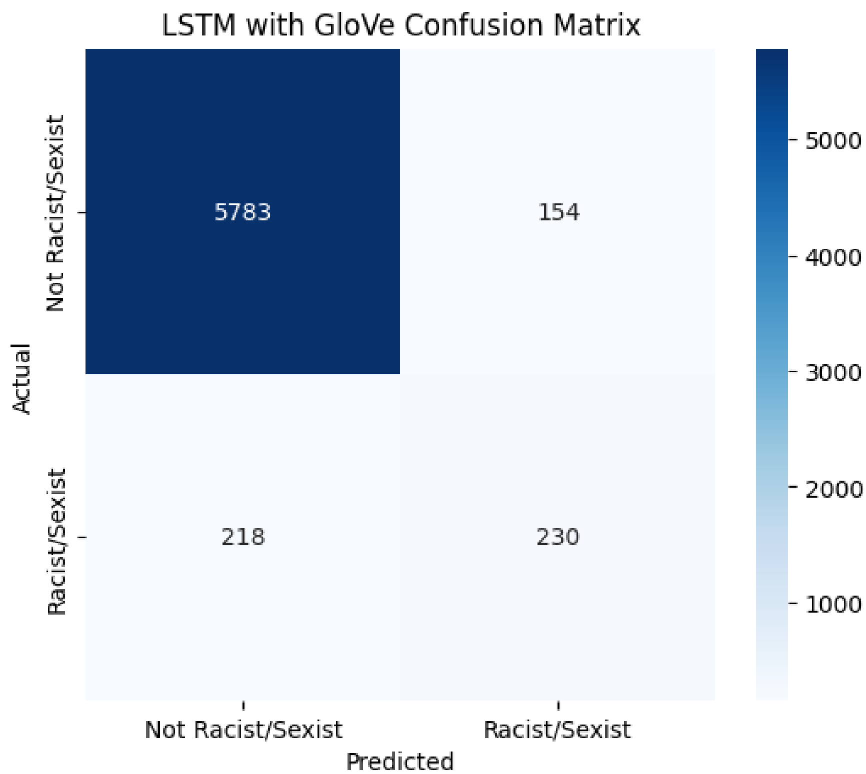

Figure 11: Confusion matrix for the LSTM with GloVe model

Tweet class	Prec.	Rec.	F1	Cases
Non-offensive class	0.96	0.98	0.97	5970
Offensive class	0.53	0.39	0.45	410
Overall accuracy			0.932	6380
Macro average	0.74	0.68	0.71	6380
Weighted average	0.93	0.93	0.93	6380

Table 4: Evaluation summary for the LSTM model using GloVe

The results show that the LSTM with GloVe is the strongest option for this dataset because it keeps overall accuracy high while improving recognition of the smaller offensive class. Nevertheless, the limited number of racist/sexist tweets remains a major challenge. Future improvements should therefore focus on balancing the dataset and selecting a decision threshold based on validation F1-score rather than depending only on the default threshold of 0.5.

5. Conclusion and Future Work

This assignment demonstrated the application of recurrent neural network models to racist and sexist tweet classification. The Simple RNN provided a useful baseline, but it was limited in detecting offensive tweets from the minority class. The LSTM improved contextual learning and increased recall for that class. The LSTM supported by pretrained GloVe embeddings achieved the strongest overall result because it combined sequential modelling with external semantic knowledge.

The main limitation of the project is the imbalance between the two labels. Since most tweets are non-offensive, a high accuracy value does not automatically mean that the model is reliable at identifying harmful content. Future work should consider oversampling or augmenting minority-class tweets, tuning the classification threshold, testing bidirectional LSTM models, adding attention mechanisms and exploring Transformer-based approaches such as BERT. A small web-based or notebook-based demonstration could also be developed so that a user can enter a tweet and receive an immediate prediction.

6. References

Hochreiter, S. and Schmidhuber, J. (1997) Long Short-Term Memory. Neural Computation, 9(8), pp. 1735-1780.

Pennington, J., Socher, R. and Manning, C.D. (2014) GloVe: Global Vectors for Word Representation. Proceedings of EMNLP 2014, pp. 1532-1543.

Mikolov, T., Chen, K., Corrado, G. and Dean, J. (2013) Efficient Estimation of Word Representations in Vector Space. arXiv preprint arXiv:1301.3781.

Chollet, F. (2018) Deep Learning with Python. Shelter Island: Manning Publications.

Bird, S., Klein, E. and Loper, E. (2009) Natural Language Processing with Python. Sebastopol: O'Reilly Media.

Pedregosa, F. et al. (2011) Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, pp. 2825-2830.